



Asset Performance: Hea 01 Daylighting



Credits



No Minimum Standard

Aim

To ensure asset users have access to good levels of daylight.

Question

Do relevant occupied spaces have glazed areas that allow asset users to experience daylight?

Credits	Answer	Select a single answer option
0	A.	Question not answered
0	B.	No
1	C.	Yes, $\geq 10\%$ of the total area of the asset's external walls and roof is glazed
2	D.	Yes, $\geq 50\%$ of occupied space meets the minimum performance requirements for glazed area as a percentage of floor area
4	E.	Yes, $\geq 80\%$ of occupied space meets the minimum performance requirements for glazed area as a percentage of floor area
Exemplary	F.	Yes, all occupied space meets the minimum performance requirements for glazed area as a percentage of floor area

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Filtering This assessment issue can be filtered out where the only relevant occupied space requires daylight to be excluded or lighting conditions to be strictly controlled.	All
2.	Glazed areas must let daylight pass directly into the asset and include side windows and roof lights.	C – F
3.	For each relevant occupied space, the glazed area expressed as a percentage of the floor area (see Methodology) meets or exceeds, the minimum performance requirements in Table 15. Credits are awarded based	D – F

Criterion	Assessment criteria	Applicable Answer
	on the percentage of all relevant occupied space (weighted by floor area) that meets the minimum performance requirements.	
4.	Only relevant occupied spaces are to be included in the assessment. Spaces that do not need to be assessed include, but are not limited to; changing rooms, break out space, meeting and conference rooms, toilet areas, and circulation space.	D – F

Methodology

Calculating glazed area as a percentage of floor area in occupied spaces and assessment against the minimum performance requirements

The following steps must be followed in order to assess whether the asset demonstrates compliance with the minimum performance requirements. Where daylight needs to be excluded from or lighting conditions need to be strictly controlled in an occupied space, e.g. media or arts production areas, special education needs sensory space or x-ray rooms, these spaces can be excluded from assessment.

1. Determine the latitude for the asset's location.
2. Calculate the net internal area (m²) for each relevant occupied space and the total net internal area (m²) for all relevant occupied spaces.
3. For each relevant occupied space, calculate the total glazed area (m²) of all windows and roof lights within the space. The frames are not included in the calculation, only the area of glass.
4. For each relevant occupied space, calculate the percentage of glazed area to floor area as follows:

$$\text{Percentage glazed area to floor area} = \left(\frac{\text{Total glazed area in occupied space}}{\text{Net internal area of occupied space}} \right) \times 100$$

5. For occupied spaces with tinted or diffusing glazing, it is necessary to apply a glass transmittance weighting to the glazed area to floor area percentage. For clear or low emissivity single, double or triple glazing, no weighting correction is required. For tinted or diffusing glazing, the glazing to floor area percentage is multiplied by T/0.8, where T is the light transmittance of the glazing (expressed as a decimal). If the glass transmittance is not known, the glazing to floor area percentage is multiplied by 0.25 for tinted glazing and by 0.60 for diffusing glazing.
6. For each relevant occupied space, check that no part of the occupied space is more than x metres on plan from the glazed area, where x is three times the height of the top of the glazed area above the floor level. For example, if the height to the top of a glazed area from the floor is 3m, then no part of the occupied space must be further than 9m from the glazed area. Where an occupied space does not meet this distance requirement, it is not possible for the space to comply with the minimum performance requirements.
7. For each relevant occupied space, check if there are any external obstructions (e.g. adjacent buildings or structures, trees or hedges, etc.) that will potentially limit the amount of daylight reaching the space. If there is an obstruction, its horizontal distance away from the glazing must be greater than its height above the centre of the glazing. For example, if an obstruction's height is 10m above the centre of the glazing, then the obstruction must be at least 10m away from the glazing. Where an obstruction does not meet this distance requirement, the obstructed glazing does not count towards the total glazed area in the calculation.

8. For each relevant occupied space, compare the transmittance weighted glazed area to floor area percentage to the minimum performance requirements in Table 15 for the asset's latitude to assess whether the occupied space is compliant or not.
9. Calculate the percentage of all relevant occupied space within the asset that meets the minimum performance requirements in accordance with steps 4 - 8 above, and therefore determine the number of credits that can be awarded, as follows:

Percentage of relevant occupied space

$$= \left(\frac{\text{Total net internal area of compliant occupied space}}{\text{Total net internal area of all relevant occupied space}} \right) \times 100$$

Daylighting calculations, measurement and simulation

Where daylighting calculations, measurement or simulation have previously been performed for the asset, for example as part of the asset's design and construction, and there have been no changes to the layout or glazing in relevant occupied space and there are no new external obstructions, then these studies may be used as an alternative method of demonstrating compliance with the criteria. The daylighting studies must demonstrate that daylight levels in $\geq 50\%$ or $\geq 80\%$ of relevant occupied space meet local good practice requirements (e.g. achievement of the BREEAM New Construction or Refurbishment and Fit-out scheme 'Daylighting' credits).

Checklists and Tables

Table 15: Minimum performance requirements for glazed area as a percentage of floor area in occupied space

Latitude	$< \pm 40^\circ$	$\geq \pm 40^\circ < \pm 45^\circ$	$\geq \pm 45^\circ < \pm 50^\circ$	$\geq \pm 50^\circ < \pm 60^\circ$	$\geq \pm 60^\circ$
Glazed area to floor area	$\geq 7\%$	$\geq 8\%$	$\geq 9\%$	$\geq 10\%$	$\geq 11\%$

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
2 - 4	Documentation specifying the floor area and glazed area in occupied spaces and calculations in accordance with the Methodology.
2 - 4	Photographic evidence of building elevations, glazed areas in occupied space and any obstructions.

Definitions

Occupied space:

A room or space within the asset that is likely to be continuously occupied for 30 minutes or more per day by an asset user.

Additional information

Glazed areas and overheating

This assessment issue focuses on the provision of minimum glazed areas for the purposes of allowing good levels of daylight within occupied space. As such, no maximum limit has been stipulated. However, overheating in a space can be a problem if glazed areas are too large.